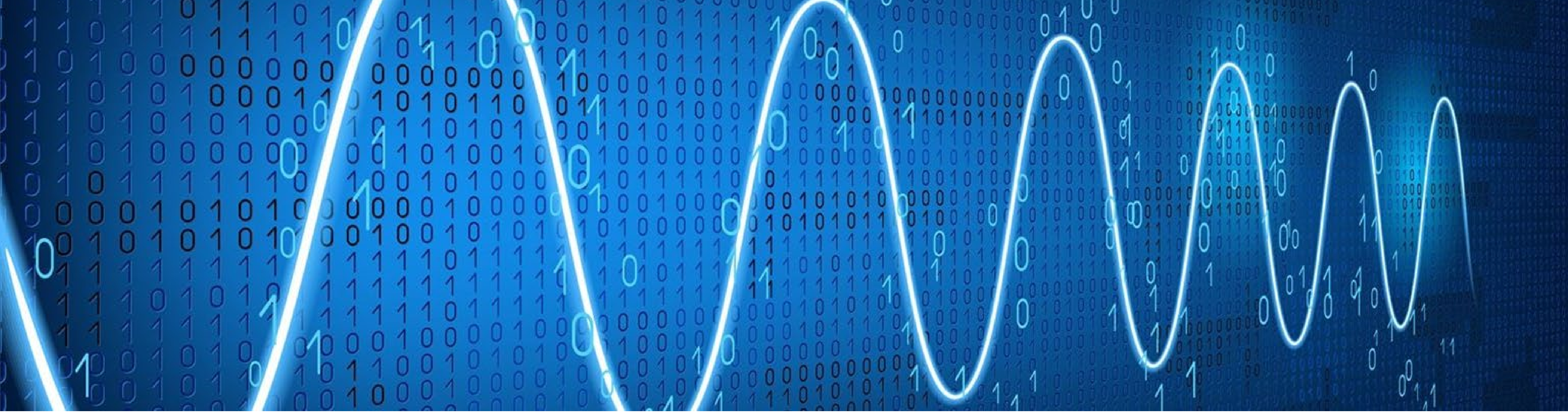

Lecture 10:
IIR Filters, Linear Phase Filters

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Where We Are in the Course

- Analog Signal Processing, Introduction to digital signal processing (Lectures 2-6)
- LTI System (Lectures 7 and 9)
- Z-Transform, System Function, Frequency Response (Lectures 8)
- FIR Filters (Lecture 8)
- ***IIR Filters, Linear Phase Filters*** (Lecture 10)
- Implementation Aspects (Lecture 11)
- CTFT, DTFT, DFT (Lecture 12)
- FFT (Lecture 13)

Lecture 14: Sample Questions

Section 8,
Signal Processing First

Outline

- How are poles and zeros of $H(z)$ influencing $\left|H(e^{j\omega})\right|$?
- Designing IIR Filters
- Linear Phase Filters

How are poles and zeros of $H(z)$ influencing $|H(e^{j\omega})|$?

Let's assume that our $H(z)$ is written in the following form:

$$H(z) = G_1 \frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}$$

where z_ks are zeros and p_ks are poles, and we want to find

$$\left|H(e^{j\omega})\right| = |H(z)|_{z=e^{j\omega}}$$

$$= |G_1| \frac{|e^{j\omega}-z_1||e^{j\omega}-z_2|\cdots|e^{j\omega}-z_M|}{|e^{j\omega}-p_1||e^{j\omega}-p_2|\cdots|e^{j\omega}-p_N|}$$

We see that all factors (in numerator and denominator)

$$\left|e^{j\omega}-z_k\right| \text{ and } \left|e^{j\omega}-p_k\right|$$

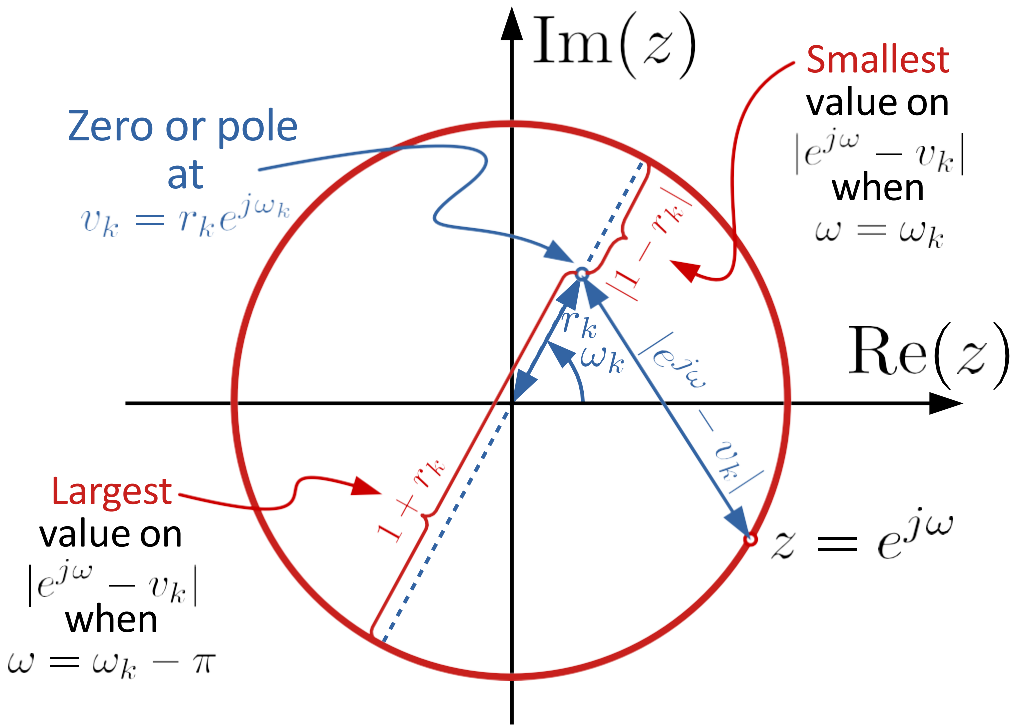
are distances from the point $z = e^{j\omega}$ to a zero or a pole With this knowledge, we can find the magnitude of $X(\omega)$ by geometric reasoning (again in the z-plane)

How are poles and zeros of H(z) influencing $\left|H(e^{j\omega})\right|$?

For $\left|H(e^{j\omega})\right|$ we have

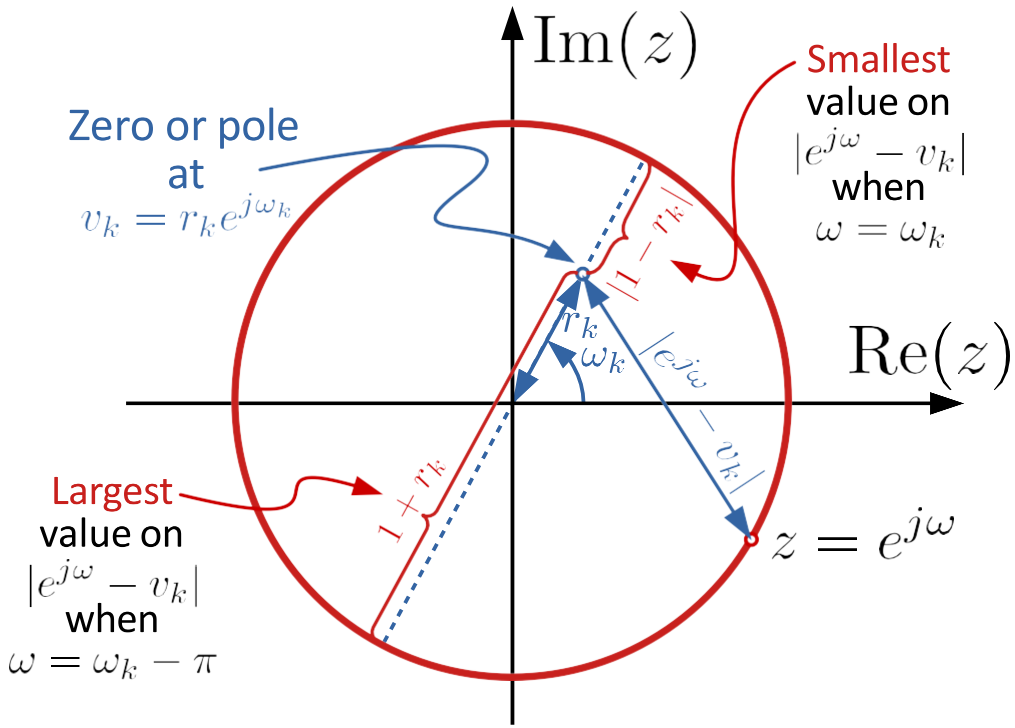
$$\left|H(e^{j\omega})\right| = |G_1| \frac{|e^{j\omega} - z_1||e^{j\omega} - z_2| \cdots |e^{j\omega} - z_M|}{|e^{j\omega} - p_1||e^{j\omega} - p_2| \cdots |e^{j\omega} - p_N|}$$

For simplicity, first a single zero or pole that we call Type equation here. v_k (since it can be either z_k and p_k) and $|e^{j\omega} - v_k|$ is of interest



How are poles and zeros of H(z) influencing $\left|H(e^{j\omega})\right|$?

$$\left|H(e^{j\omega})\right| = |G_1| \frac{|e^{j\omega} - z_1||e^{j\omega} - z_2| \cdots |e^{j\omega} - z_M|}{|e^{j\omega} - p_1||e^{j\omega} - p_2| \cdots |e^{j\omega} - p_N|}$$



If $v_k = z_k$ is a zero, $|e^{j\omega} - z_k|$ is in the numerator and will contribute with a factor in the range

$$|1 - |z_k|| \leq |e^{j\omega} - z_k| \leq 1 + |z_k|$$

Equality if $\omega = \angle z_k$

Equality if $\omega = \angle z_k - \pi$

If $v_k = p_k$ is a pole, $|e^{j\omega} - p_k|$ is in the denominator and will contribute with a factor in the range

$$\frac{1}{1 + |p_k|} \leq \frac{1}{|e^{j\omega} - p_k|} \leq \frac{1}{|1 - |p_k||}$$

Equality if $\omega = \angle p_k - \pi$

Equality if $\omega = \angle p_k$

Two “simple” conclusions:

A zero close to the unit circle will attenuate $\left|H(e^{j\omega})\right|$ near its location (at angular frequencies near its phase). The closer the stronger the attenuation (at unit circle: $\left|H(e^{j\omega})\right| = 0$)

A pole close to the unit circle will amplify $\left|H(e^{j\omega})\right|$ near its location. The closer the stronger the amplification

Example:

Draw the pole-zero locations for $H(z)$ and try to figure out where $|H(e^{j\omega})|$ is large and small, when

$$H(z) = \frac{1 - z^{-1}}{1 - z^{-1} + \frac{1}{2} z^{-2}}$$

Example:

Draw the pole-zero locations for $H(z)$ and try to figure out where $\left|H(e^{j\omega})\right|$ is large and small, when

$$H(z) = \frac{1 - z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

First, break out the highest powers of z^{-1} in numerator and denominator

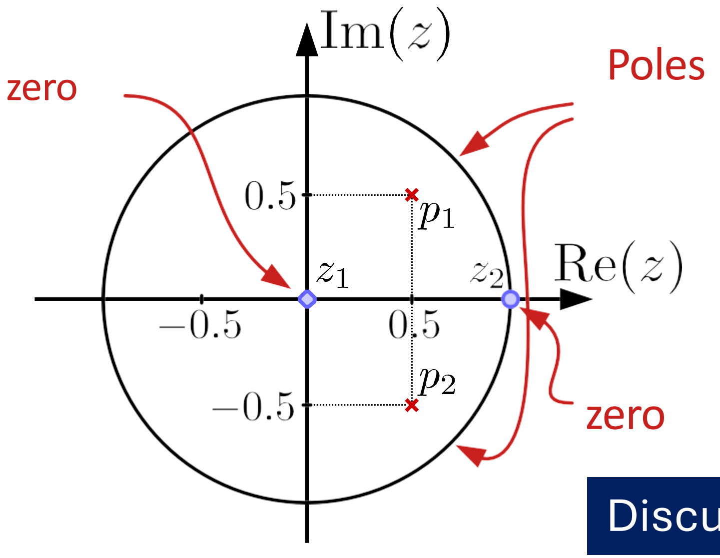
$$H(z) = \frac{z^{-1}}{z^{-2}} \frac{z - 1}{z^2 - z + \frac{1}{2}}$$

$$= z\frac{z - 1}{z^2 - z + \frac{1}{2}}$$

The first factor contributes with a zero at $z_1 = 0$ and the numerator of the fraction another one at $z_2 = 1$

Finding the roots of the second degree polynomial in the denominator gives two conjugate poles at $p_1 = 0.5(1+j)$ and $p_2 = 0.5(1-j)$

Pole-zero diagram



Discussion Time I

A zero close to the unit circle will attenuate $\left|H(e^{j\omega})\right|$ near its location (at angular frequencies near its phase). The closer the stronger the attenuation (at unit circle: $\left|H(e^{j\omega})\right| = 0$)

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Example:

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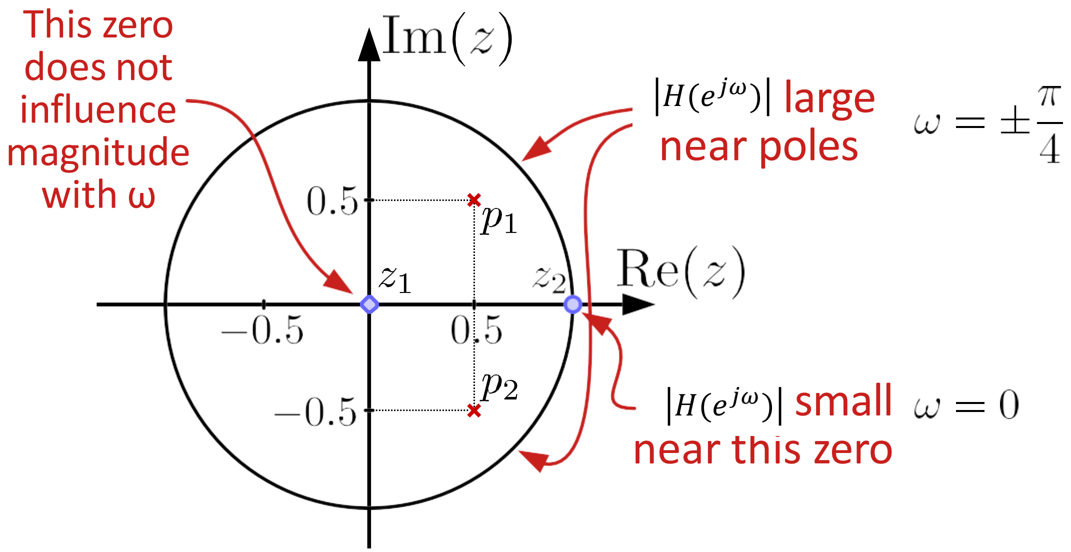
Pole-zero diagram



Example:

Draw the pole-zero locations for $H(z)$ and try to figure out where $\left|H(e^{j\omega})\right|$ is large and small, when

$$H(z) = \frac{1 - z^{-1}}{1 - z^{-1} + \frac{1}{2} z^{-2}}$$

First, break out the highest powers of z^{-1} in numerator and denominator

$$H(z) = \frac{z^{-1}}{z^{-2}} \frac{z - 1}{z^2 - z + \frac{1}{2}}$$

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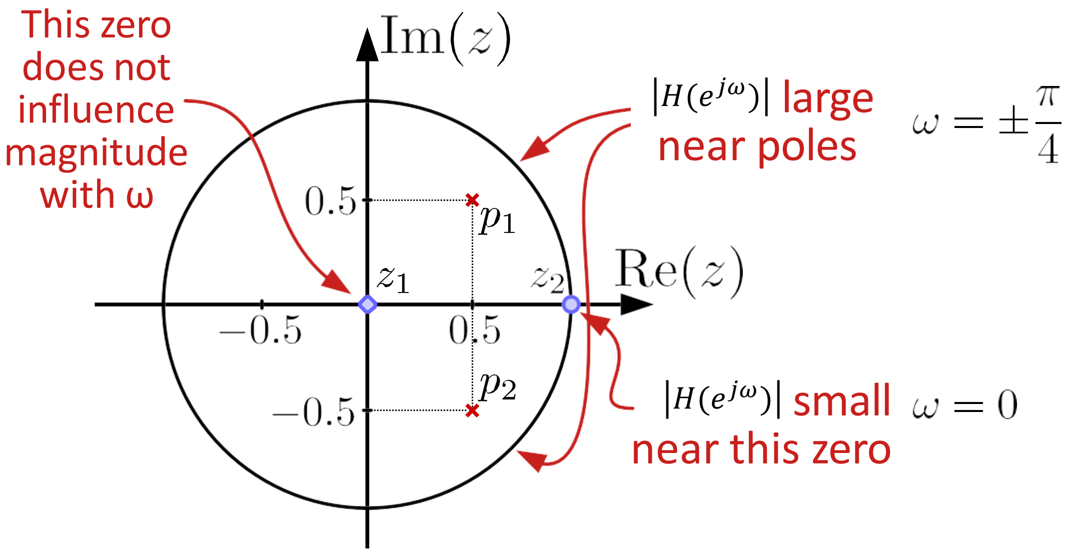
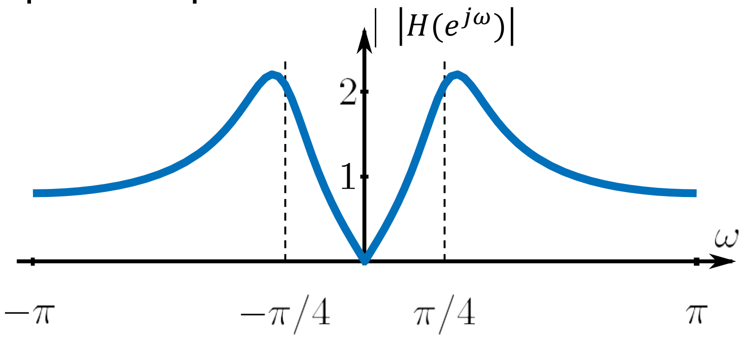
Pole-zero diagram



Let's plot $\left|H(e^{j\omega})\right|$ and see ...



Designing IIR Filters:

-Influence of the poles and zeros of $H(z)$ influencing $\left|H(e^{j\omega})\right|$

-Causality and Stability (All Poles inside the Unit Circle)

- Real-Value Coefficient (Conjugate-Pair Zeros and Poles)

Discussion Time II

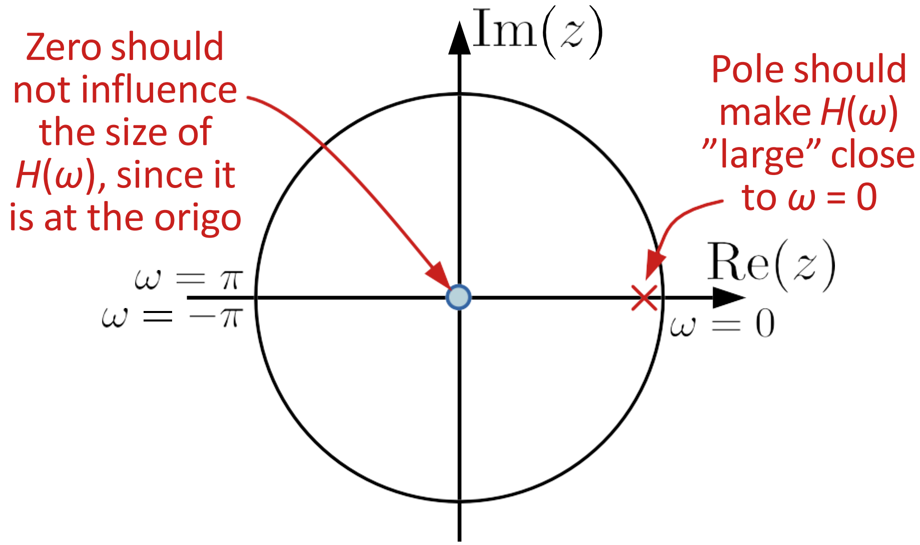
To create a low-pass (LP) filter, where should a single pole be placed to allow only low-frequency components to pass through, considering the pole's angular frequency?

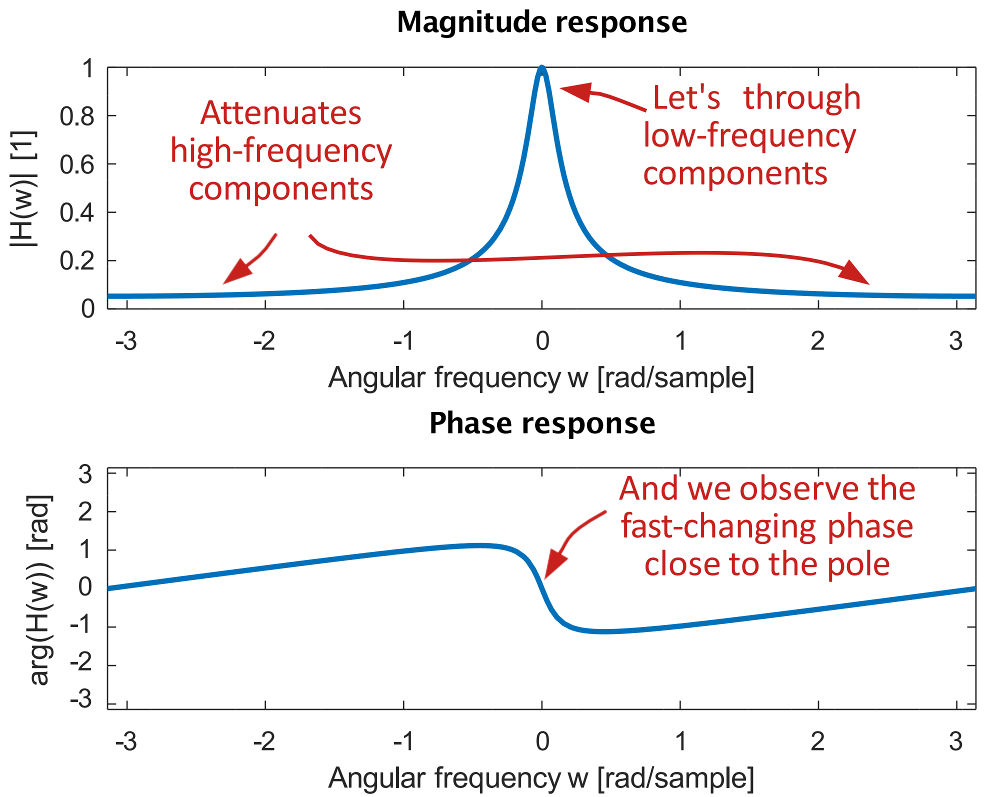
Example:

$$H(z) = \frac{1 - 0.9}{1 - 0.9z^{-1}} = \frac{0.1z}{z - 0.9}$$

shows that there is also a zero at z = 0

The pole-zero plot becomes



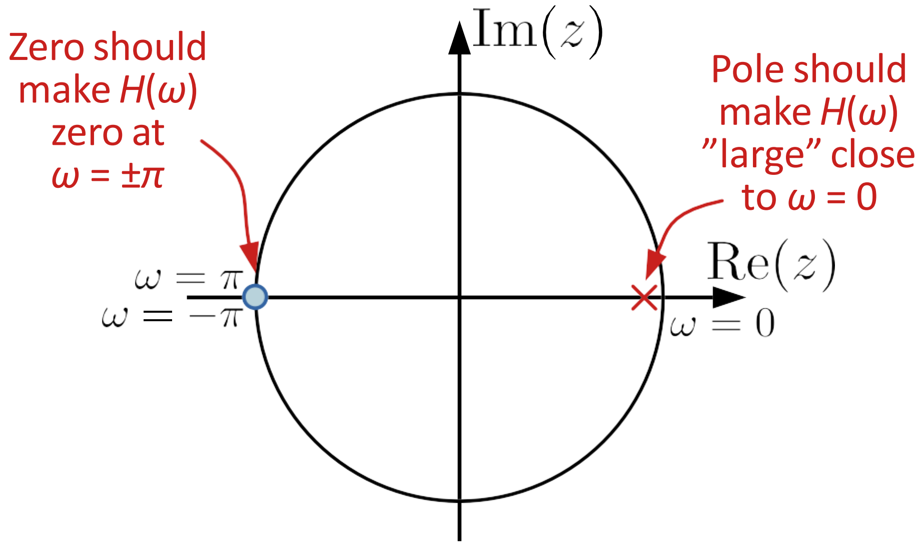


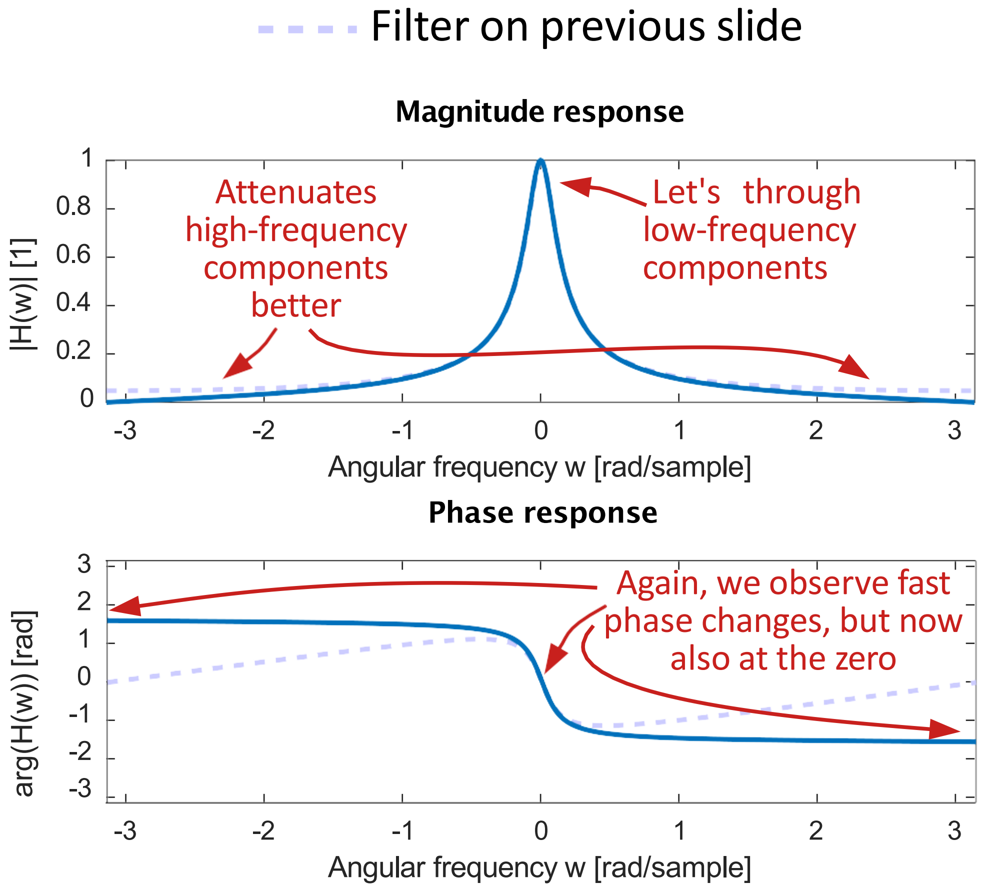
Not great attenuation at $\omega = \pm\pi$

Example:

$$H(z) = \frac{0.05(1+z^{-1})}{1-0.9z^{-1}} = \frac{0.05(z+1)}{z-0.9}$$

The pole-zero plot becomes





Attenuation at $\omega = \pm\pi$ is now “prefect”

How do we make a HP filter?

Discussion Time III

Which system function represent a better high-pass filter with only one pole?

1. System function has a pole at z=0.9 and a zero at z=0

2. System function has a pole at z=-0.9 and a zero at z=0

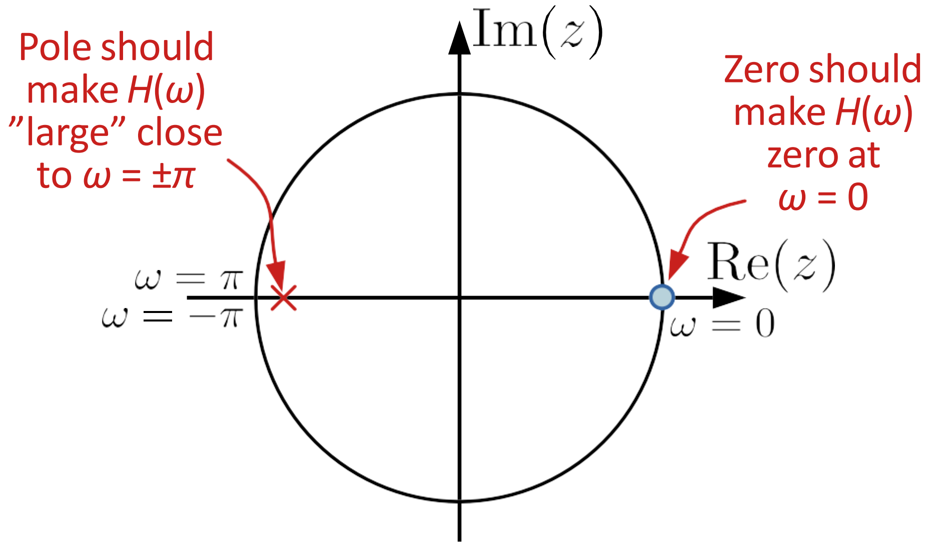
3. System function has a pole at z=0.9 and a zero at z=-1

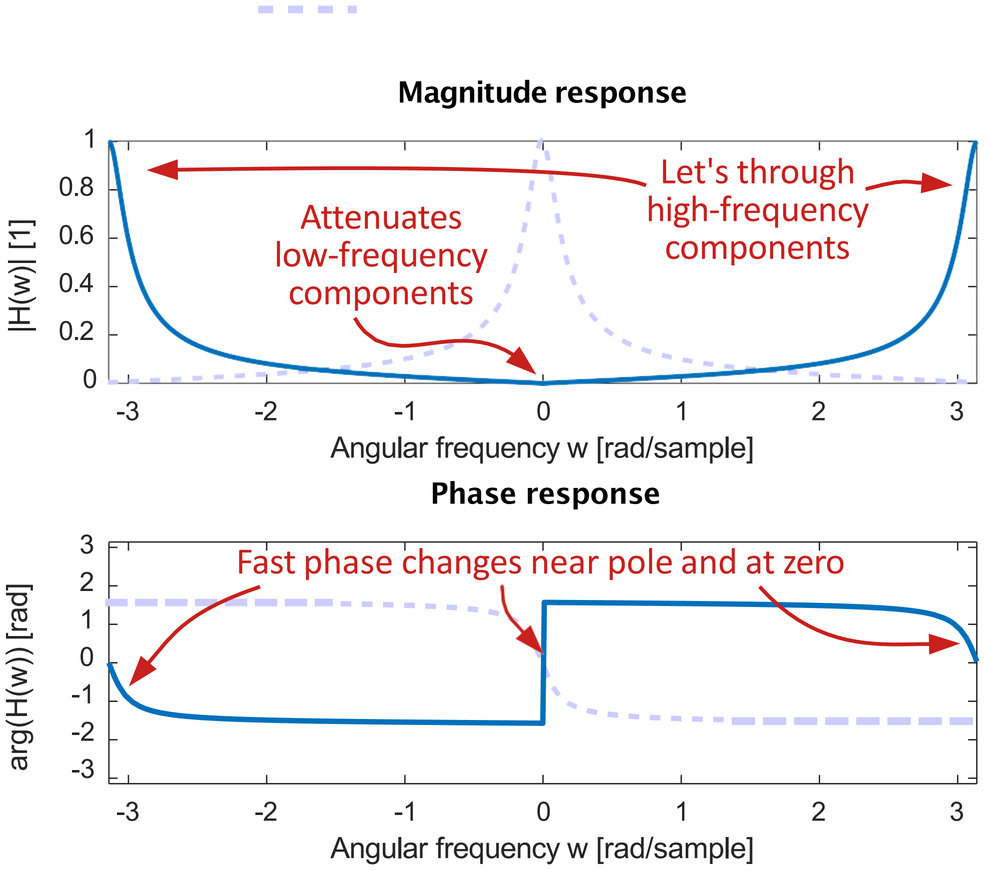
4. System function has a pole at z=-0.9 and a zero at z=1

Example: Making HP from the LP filter

Mirroring the pole and zero of the LP filter in the imaginary axis (change their real-part sign) we get

$$H(z) = \frac{0.05(1 - z^{-1})}{1 + 0.9z^{-1}} = \frac{0.05(z - 1)}{z + 0.9}$$

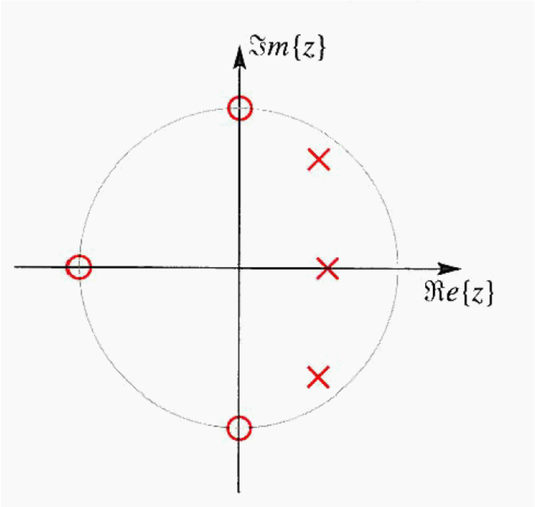


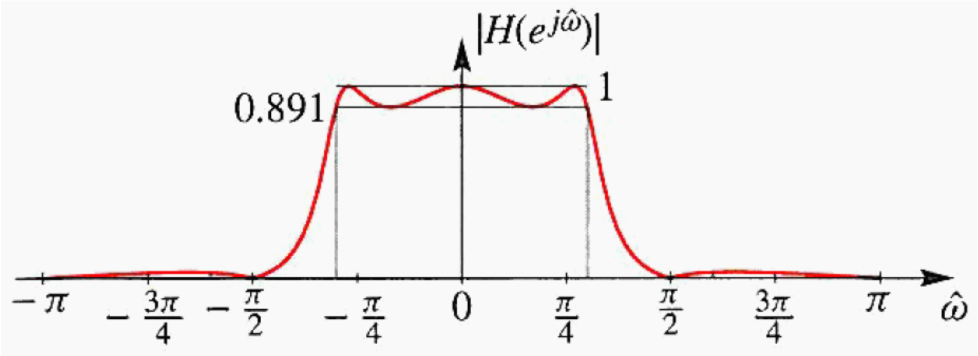


Components at $\omega = \pm\pi$ go through unchanged and attenuation at $\omega = 0$ is "prefect"

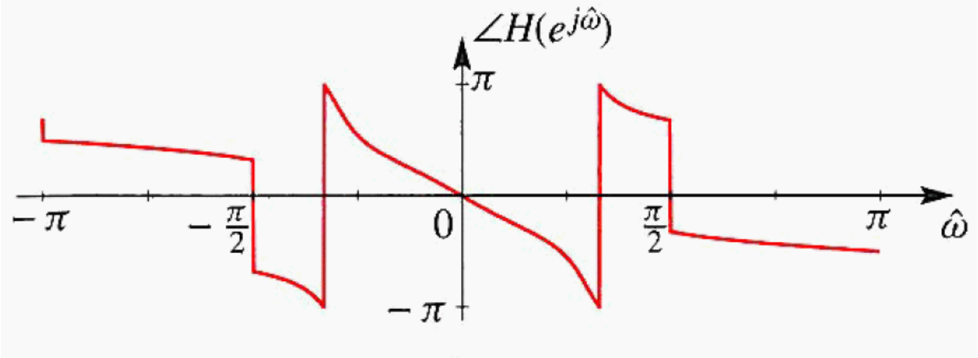
Example of a Third Order IIR Filter

$$
\begin{aligned}
H(z) &= \frac{0.0798(1+z^{-1}+z^{-2}+z^{-3})}{1-1.556z^{-1}+1.272z^{-2}-0.398z^{-3}} \\
&= \frac{0.0798(1+z^{-1})(1-e^{j\pi/2}z^{-1})(1-e^{-j\pi/2}z^{-1})}{(1-0.556z^{-1})(1-0.846e^{j0.3\pi}z^{-1})(1-0.846e^{-j0.3\pi}z^{-1})} \\
&= -\frac{1}{5} + \frac{0.62}{1-.556z^{-1}} + \frac{0.17e^{j0.96\pi}}{1-.846e^{j0.3\pi}z^{-1}} + \frac{0.17e^{-j0.96\pi}}{1-.846e^{-j0.3\pi}z^{-1}}
\end{aligned}
$$





(a)



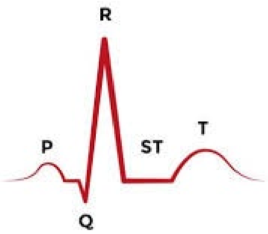
(b)

Linear Phase Filters

Linear Phase Filters

Reminder

$x[n] = A\cos(\omega_0 n)$ → $H(e^{j\omega})$ → $y[n] = A\,\left|H(e^{j\omega_0})\right|\cos(\omega_0 n + \measuredangle H(e^{j\omega_0}))$



$x[n] = \sum_{k=1}^{L} A_k\cos(\omega_k n)$ → $H(e^{j\omega})$ → $y[n] = \sum_{k=1}^{L} A_k\left|H(e^{j\omega_k})\right|\cos(\omega_k n + \measuredangle H(e^{j\omega_k}))$

$= \sum_{k=1}^{L} A_k\left|H(e^{j\omega_k})\right|\cos(\omega_k(n + \measuredangle H(e^{j\omega_k})/\omega_k)$

N[n]+$x[n]$ → $H(e^{j\omega})$ → $y[n] = \sum_{k=1}^{L} A_k\left|H(e^{j\omega_k})\right|\cos(\omega_k(n + \measuredangle H(e^{j\omega_k})/\omega_k)$

Linear Phase Filters

When we apply a filter to an input signal, it is often beneficial if all frequency components experience the same delay (and are not shifted relative to each) so that the shape of the signal is maintained.

If we want all frequency components to experience the same delay, the requirement is that $\Delta(\omega)$ is constant for all ω
Assuming that constant delay is Δ_0, we can write the requirement as

$$\Delta_0 = -\measuredangle H(e^{j\omega})/\omega \Rightarrow \measuredangle H(e^{j\omega}) = \boxed{\underbrace{-\Delta_0\omega}_{\text{Linear function of } \omega}}$$

and we see that the phase response of the filter must be a linear function of ω (with slope $-\Delta_0$)

Linear Phase Filters

Which filters are linear-phase filters?

For FIR filters (of length $N+1$) with real-valued coefficients, there are three types

1) Non-causal with symmetry around $n = 0$

$$h(n) = h(-n)$$

for which $H(\omega)$ is real-valued (phase 0 or $\pm\pi$)

2) Causal with symmetry around $n = N/2$

$$h(n) = h(N-n)$$

3) Causal with anti-symmetry around $n = N/2$

$$h(n) = -h(N-n)$$

Linear Phase Filters

The symmetry conditions on impulse responses can be expressed in terms of the z-transform $H(z)$ [using the time-reversal and delay properties, Lecture 2, slide 14]

1) Non-causal with symmetry around n = 0

$$h(n) = h(-n) \overset{z}{\longleftrightarrow} H(z) = H(1/z)$$

2) Causal with symmetry around n = N/2

$$h(n) = h(N-n) \overset{z}{\longleftrightarrow} H(z) = z^{-N} H(1/z)$$

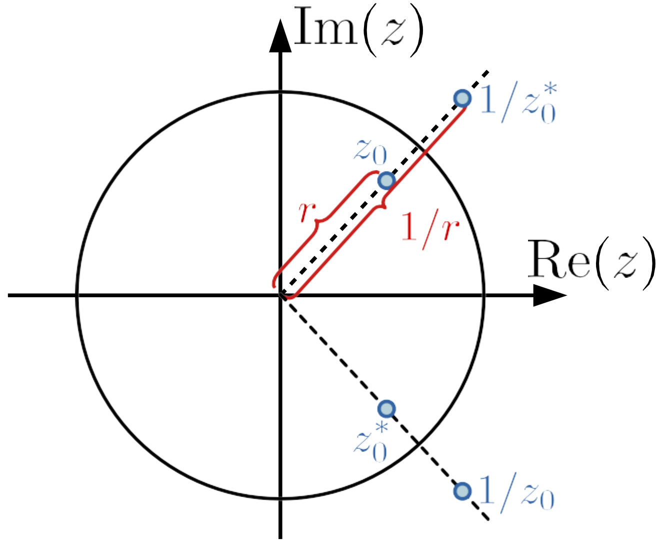
3) Causal with anti-symmetry around n = N/2

$$h(n) = -h(N-n) \overset{z}{\longleftrightarrow} H(z) = -z^{-N} H(1/z)$$

What they have in common is that if z_0 is a zero of $H(z)$, then its reciprocal $z_1 = 1/z_0$ must also be a zero of H(z)

For real impulse responses we also know that the zeros must come in conjugate pairs

Location of zeros for linear-phase, real-valued, FIR filters



We see that linear phase requires each zero inside the unit circle has a corresponding one outside the unit circle

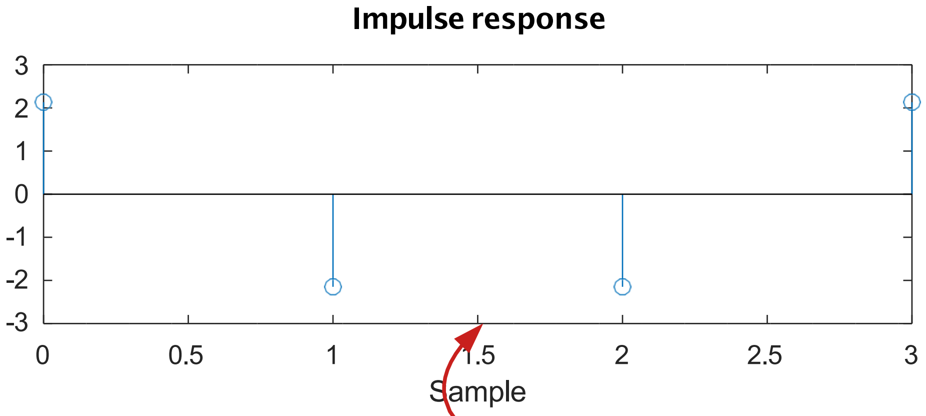
Linear Phase Filters

How about Infinite Impulse Response (IIR) filters?

The same reasoning can be done for poles, but requiring that poles inside the unit circle also have corresponding ones outside would make the IIR filter unstable … so BIBO stable IIR filters cannot have linear phase

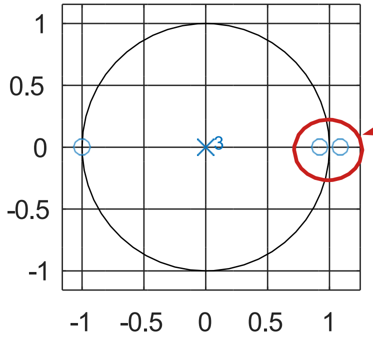
Example:

Causal impulse response of duration 4

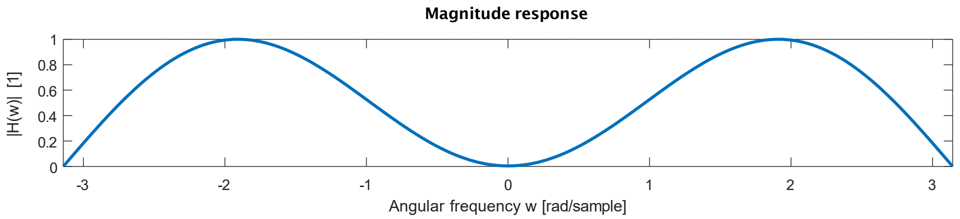


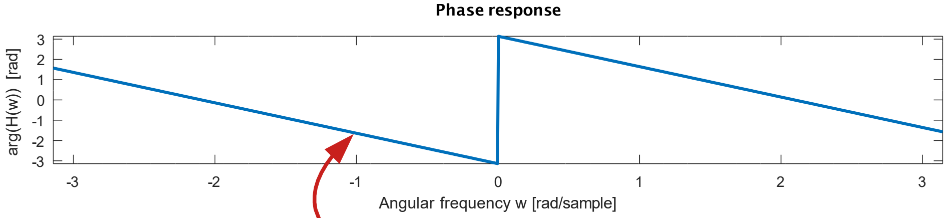
Symmetric around this point

Pole-zero plot



Reciprocal zeros on real axis





Linear phase